Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai: 400058, India
(Autonomous College of Affiliated to University of Mumbai)

End Semester Examination
Dec.2024

Max. Marks: 100 | **Duration: 3 Hrs**

Class: TE | **Semester: V**

Course Code:CS301/AI301/DS301 | **Program: CE/CSE**

Name of the Course: Theory of Computation

Instructions:
(1) All questions are compulsory.
(2) Draw neat diagrams and write equations wherever necessary.
(3) Assume suitable data if necessary

Course Outcomes: At the end of the course student will be able to:
(1) Design finite automaton for regular expressions and languages.
(2) Apply the properties of regular languages.
(3) Construct the grammar for a language and convert it into normal forms.
(4) Design and evaluate pushdown automata and Turing Machine for a language.

Q No.		Max. Marks	CO BL PI
Q.1a	Give examples of two languages L_1 and L_2 which are regular. Also state true or false with appropriate justification for the following statement: There are two languages L_1 and L_2 such that L_1 and L_2 are not regular but $L_1 \cap L_2$ is.	10	2-2
	OR		
	Give examples of two languages L_1 and L_2 which are not regular. Also state true or false with appropriate justification for the following statement: There are two languages L_1 and L_2 such that L_1 and L_2 are not regular but $L_1 \cdot L_2$ is.	10	2-2
Q.1b	Prove or disprove the following statement: Let L_1 be a regular language and L_2 be a non-regular language and $L_1 \cap L_2 = \emptyset$ where $\emptyset$ is the empty set. Then $L_1 \cup L_2$ can never be regular.	10	2-3

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<table>
<tr><td>Q.2a</td><td>Reduce the following production of grammar to Greibach Normal Form
S → ABA
A → aA | ε
B→ bB | ε</td><td>10</td><td>3-3</td></tr>
<tr><td>Q2.b</td><td>Let L, L_1 and L_2 be languages over the alphabet $\Sigma = \{a, b\}$.
Prove/disprove the following statements:
(a) $L^*L^* = L^*$
(b) If $L_1 \cdot L_2 = L_2 \cdot L_1$ then $L_1 = L_2$
(c) $(L^*)^* = L$</td><td>10</td><td>2-2</td></tr>
<tr><td>Q.3a</td><td>Construct a CFG for the language consisting of all strings of a's and b's in which b never comes before a.</td><td>10</td><td>3-3</td></tr>
<tr><td>Q.3b</td><td>Prove undecidability of halting problem.</td><td>10</td><td>4-3</td></tr>
<tr><td>Q.4a</td><td>Minimize the following Deterministic finite automata:
<table><tr><th>Q\ Σ</th><th>a</th><th>b</th></tr><tr><td>Q0</td><td>Q1</td><td>Q3</td></tr><tr><td>Q1</td><td>Q0</td><td>Q3</td></tr><tr><td>Q2</td><td>Q1</td><td>Q4</td></tr><tr><td>Q3*</td><td>Q5</td><td>Q5</td></tr><tr><td>Q4</td><td>Q3</td><td>Q3</td></tr><tr><td>Q5*</td><td>Q5</td><td>Q5</td></tr></table>Note: Q3 and Q5 are final states, Q0 is initial state.</td><td>10</td><td>1-3</td></tr>
<tr><td>Q.4b</td><td>Examine how the four levels of the Chomsky hierarchy relate to one another with neat, labelled diagram and appropriate examples.</td><td>10</td><td>3-3</td></tr>
<tr><td>Q.5a</td><td>Design a Turing machine to perform m * n (* is product of m and n) such that:
i/p: $B0^m10^nB$ (where B represents blank space, and m,n>1)
o/p: $B0^{m*n}B$

OR
Design a Turing machine to copy given number 'n' on the tape such that:
i/p: $B0^nB$ (where B represents blank space, and m,n>1)
o/p: $B0^n10^nB$</td><td>10

10</td><td>4-3

4-3</td></tr>
<tr><td>Q.5b</td><td>Design a push down automata to recognise equal numbers of a's and b's over the input (a, b)*</td><td>10</td><td>4-3</td></tr>
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